

National Forensic Sciences University

School of Cybersecurity and Digital Forensics

M.Sc. DFSI - Semester I

MSDF26-PC-103 Programming for Digital Forensics

Lab Assignment #1: Python Fundamentals, Mathematical Operations and Control Structures

Objective

1. Understand the basic syntax and structure of a Python program.
2. Use variables and appropriate data types to store and process user-supplied data.
3. Apply input and output operations using the input() and print() functions.
4. Perform basic mathematical operations using arithmetic, comparison, and logical operators.
5. Apply decision-making statements such as if, if-else, and if-elif-else to solve programming problems.
6. Use nested conditional statements to evaluate multiple conditions.
7. Validate user input and handle special conditions, such as division by zero and invalid marks.
8. Apply the while loop to perform repetitive operations and process the digits of a number.

Sr No	Description
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Task-1	Basic Arithmetic Calculator
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Write a Python program that accepts two numbers from the user and displays their sum, difference, product, quotient, and remainder. Handle division by zero using an appropriate conditional statement.

Task-2	Area and Perimeter Calculator
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Write a program that accepts the length and width of a rectangle and calculates its area and perimeter. Also, determine whether the given dimensions form a square.

Task-3	Temperature Converter
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Write a program that accepts a temperature in Celsius and converts it into Fahrenheit. Display whether the temperature is below freezing, normal, or hot.

Task-4	Even, Odd, and Divisibility Check
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Write a program to accept an integer and determine whether it is even or odd. Also check whether it is divisible by 5 and 7.

Task-5	Largest of Three Numbers
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Write a Python program that accepts three numbers and displays the largest number.
If all three numbers are equal, display an appropriate message.

Task-6 Student Grade Calculator

Accept marks obtained in five subjects, each out of 100. Calculate the total, percentage, and grade using the following criteria:

- 90% or above: Grade A+
- 80%–89%: Grade A
- 70%–79%: Grade B
- 60%–69%: Grade C
- 50%–59%: Grade D
- Below 50%: Fail

Validate that every mark is between 0 and 100.

Task-7 Electricity Bill Calculation

Write a program to calculate an electricity bill based on the number of units consumed:

- First 100 units: ₹2 per unit
- Next 100 units: ₹3 per unit
- Next 100 units: ₹5 per unit
- Above 300 units: ₹7 per unit

Add a fixed charge of ₹100 to the final bill.

Task-8 Sum and Reverse of Digits

Write a program that accepts a positive integer and uses a while loop to:

- Calculate the sum of its digits
- Count the number of digits
- Display the number in reverse order

Marking Rubric:

Total Marks: 40, Each task carries 5 marks.

Assessment criterion	Marks per task
Understanding of the problem and selection of correct logic	1
Correct use of variables, operators and control structures	1
Program correctness and expected output	2
Code readability, indentation and meaningful variable names	0.5
Testing with appropriate input values	0.5
Total per task	5